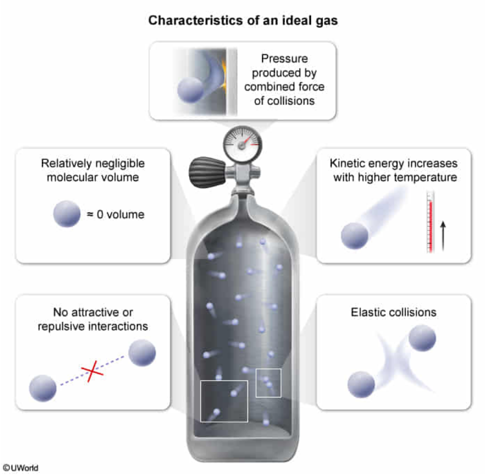
Which of the following statements describe a characteristic of an ideal gas?

I. There are no attractive or repulsive forces between the gas molecules.
II. The size (molecular volume) of the individual gas molecules is negligible.
III. Collisions between the gas molecules are completely elastic.

- A. I and II only [13%]
- B. I and III only [7%]
- C. II and III only [6%]
- ✔ D. I, II, and III [72%]

Correct
72% answered correctly

Explanation:



Sometimes when describing chemical systems that have many complicated interactions, it is useful to design models built on simplifying assumptions. The assumptions chosen for a model are often made to approximate or ignore certain effects within the system. If the chosen assumptions work reasonably well, the model built on them will give a good approximate description of the system.

One such model is that of the **ideal gas**. An ideal gas is a hypothetical gas with characteristics that are consistent with certain simplifying assumptions made for systems involving real gases. An ideal gas has four characteristics:

- An ideal gas has **no attractive or repulsive forces** between the gas molecules **(Number I)**.
- The size (**molecular volume**) of the individual gas molecules of an ideal gas is **negligible** (taken to be zero) compared to the volume (space) of the container the gas occupies **(Number II)**.
- **Collisions** between the molecules of an ideal gas are completely elastic (no energy is lost by interactions or friction) **(Number III)**.
- Ideal gas molecules have an average **kinetic energy** (energy of motion) that is directly proportional to the **gas temperature**.

Conceptually, these ideal characteristics assume that all of the kinetic energy of ideal gas molecules manifests in container collisions to produce pressure and that none of the kinetic energy is lost due to other interactions. Systems involving real gases with minimal molecular interactions can be described reasonably well by this model.

Educational objective:

An ideal gas is a hypothetical gas consisting of molecules that have no intermolecular attractions or repulsions, no molecular volume, perfectly elastic collisions, and an average kinetic energy that is directly proportional to the gas temperature. The ideal gas concept is used as a model for systems involving real gases.

Time spent:0 sec	**QID**:402194
Subject:	**System:**
General Chemistry	Thermodynamics, Kinetics & Gas Laws